

Short summary of estimation revisions

Changes on codex/rulof_new_work relative to efficient

26 August 2026

Scope

This note summarises the paper-facing results produced on the branch. It is organised by the corresponding section and table in the paper. Technical implementation details are retained in the module READMEs and estimation scripts; this note focuses on changes that affect estimates or interpretation. Across the main tables, shared rates appear in the same order: entry, exit, misclassification, and employment or initial employment. Model-specific error or duration rates follow those common quantities.

1. Baseline transition model: Table 1 and Appendix Table A1

- Re-estimated the three-wave model using the exact eight-history likelihood.
- Repaired stationarity; used 12 reproducible starts (including nested-model starts); and required optimizer convergence, a small score, positive observed information, and valid likelihood nesting before saving results.
- Replaced stale inference with analytical survey-weighted sandwich and delta standard errors.
- Separated the free initial employment probability from the model-implied long-run employment share.

Table 1: Baseline latent-state model: implied probabilities (%)

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
Entry rate (%)	8.18	8.28	2.86	2.98	2.86	2.98
	(0.04)	(0.05)	(0.05)	(0.06)	(0.05)	(0.06)
Exit rate (%)	8.64	8.54	3.04	2.91	3.04	2.91
	(0.04)	(0.05)	(0.05)	(0.06)	(0.05)	(0.06)
Misclassification rate (%)	—	—	2.99	2.99	—	—
			(0.03)	(0.03)		
False-positive rate (%)	—	—	—	—	3.05	3.05
					(0.04)	(0.04)
False-negative rate (%)	—	—	—	—	2.93	2.92
					(0.04)	(0.04)
Employment rate (%)	48.63	49.23	48.50	50.57	48.46	50.58
	(0.09)	(0.18)	(0.09)	(0.53)	(0.09)	(0.53)
Log-likelihood	-357.208M	-357.204M	-349.378M	-349.371M	-349.376M	-349.368M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Rates are percentages. Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix and the delta method. Employment is the model-implied steady-state share. The asymmetric specifications report false-positive and false-negative rates separately. The calculation does not incorporate strata or PSU clustering.

Appendix Table A1: Baseline latent-state model: parameter estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
<i>Transition parameters</i>						
θ_0 (entry)	0.0818	0.0828	0.0286	0.0298	0.0286	0.0298
	(0.0004)	(0.0005)	(0.0005)	(0.0006)	(0.0005)	(0.0006)
θ_1 (persistence)	0.9136	0.9146	0.9696	0.9709	0.9696	0.9709
	(0.0004)	(0.0005)	(0.0005)	(0.0006)	(0.0005)	(0.0006)
α (initial empl.)	0.4863	0.4852	0.4850	0.4837	0.4846	0.4832
	(0.0009)	(0.0009)	(0.0009)	(0.0010)	(0.0009)	(0.0010)
<i>Misclassification</i>						
π (symmetric)	—	—	0.0299	0.0299	—	—
			(0.0003)	(0.0003)		
π_0 (false positive)	—	—	—	—	0.0305	0.0305
					(0.0004)	(0.0004)
π_1 (false negative)	—	—	—	—	0.0293	0.0292
					(0.0004)	(0.0004)
Log-likelihood	-357.208M	-357.204M	-349.378M	-349.371M	-349.376M	-349.368M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix. α is the initial employment probability; in stationary columns it is derived from the transition parameters. The calculation does not incorporate strata or PSU clustering.

Interpretation

- The no-error model implies quarterly entry and exit of about 8.2–8.6%.
- Allowing symmetric error lowers both rates to about 3%, with a per-wave misclassification probability of 2.99%.
- False-positive and false-negative probabilities are very similar, so asymmetry adds little to the substantive result.
- The repaired likelihood strengthens the paper’s central conclusion: relatively infrequent classification error can create a large upward bias in measured labour-market turnover.

To do

- Report likelihood-ratio evidence for stationarity and asymmetric error, and check whether transition probabilities vary across wave pairs.
- Confirm the analytical inference with a person-level bootstrap or likelihood profiles for the misclassification and transition rates.

2. Panel-matching robustness: Table 2

- Applied the same exact, non-stationary estimator, 12-start search, and convergence and curvature checks to panels A, B, and C.
- Held all sample filters fixed apart from progressively stricter matching.
- Added consistent analytical standard errors to every panel comparison.

Table 2: Sensitivity to alternative panel-matching rules

	Baseline (A)		Stricter (B)		Strictest (C)	
	No error	Symmetric error	No error	Symmetric error	No error	Symmetric error
Entry rate (%)	8.28 (0.05)	2.98 (0.06)	7.59 (0.05)	2.81 (0.06)	6.56 (0.05)	2.63 (0.06)
Exit rate (%)	8.54 (0.05)	2.91 (0.06)	7.55 (0.05)	2.48 (0.06)	6.03 (0.05)	1.93 (0.06)
Misclassification rate (%)	—	2.99 (0.03)	—	2.67 (0.03)	—	2.15 (0.03)
Initial employment rate (%)	48.52 (0.09)	48.37 (0.10)	48.52 (0.09)	48.40 (0.10)	48.94 (0.11)	48.85 (0.11)
Employment rate (%)	49.23 (0.18)	50.57 (0.53)	50.11 (0.20)	53.10 (0.58)	52.13 (0.25)	57.64 (0.70)
Log-likelihood (millions)	-357.204	-349.371	-324.427	-317.258	-243.918	-238.714
N	402,923	402,923	377,132	377,132	300,200	300,200

Note: All models estimate the initial employment probability freely; stationarity is not imposed. Long-run employment is $\theta_0/(\theta_0 + 1 - \theta_1)$. Parentheses contain individual-level survey-weighted sandwich/delta standard errors. Panels A, B, and C apply successively stricter matching rules and use otherwise identical sample filters.

Interpretation

- Stricter matching lowers both observed and corrected transition rates.
- The estimated misclassification probability declines from 2.99% in panel A to 2.15% in panel C.
- Corrected entry and exit remain much lower than no-error rates under every matching rule.

- Matching quality is therefore part of the inconsistency mechanism, but the main correction is not driven solely by the baseline matching rule.

To do

- Compare observable composition and attrition across panels A–C, not only their parameter estimates.
- Check whether the Table 4 and age/education-consistency results are also stable under the stricter matching rules.

3. Age and education consistency: Table 3 and Appendix Table A2

- Re-estimated both the indicator-only model and the model with additional effects for large age and education increments, under stationarity and with a free initial employment probability.
- Repaired the EM update, used multiple starts and likelihood checks, and added analytical standard errors.
- Interpreted age and education inconsistencies neutrally: they may arise from response error, matching error, or both.
- Added robustness allowing true transition rates to differ between records with and without an inconsistency and allowing misclassification to depend on inconsistency severity.
- Added stationary and free-initial-employment specifications in which the misclassification rate also depends on person-wave indicators for membership in matching panel B but not C and panel A but not B, defined using the Table 2 sample-construction rules.

Table 3: Age- and education-inconsistency models

	(1) Indicators only	(2) Indicators only	(3) Indicators + large increments	(4) Indicators + large increments	(5) Plus matching indicators	(6) Plus matching indicators
	Stationary	Free α	Stationary	Free α	Stationary	Free α
Entry rate (%)	2.32** (0.05)	2.60** (0.05)	2.31** (0.05)	2.60** (0.05)	2.28** (0.04)	2.57** (0.05)
Exit rate (%)	2.52** (0.05)	2.20** (0.06)	2.52** (0.05)	2.19** (0.06)	2.47** (0.05)	2.14** (0.06)
Misclassification rate (%)	3.35** (0.03)	3.35** (0.03)	3.36** (0.03)	3.36** (0.03)	3.38** (0.03)	3.38** (0.03)
Initial employment rate (%)	47.92** (0.09)	47.60** (0.10)	47.92** (0.09)	47.60** (0.10)	47.92** (0.09)	47.59** (0.10)
Log-likelihood	-333.651M	-333.601M	-333.352M	-333.301M	-332.881M	-332.826M
N	394,823	394,823	394,823	394,823	394,823	394,823
Stationarity imposed	Yes	No	Yes	No	Yes	No
Large-increment effects	No	No	Yes	Yes	Yes	Yes
Matching-quality indicators	No	No	No	No	Yes	Yes

Note: Entry, exit, and mean misclassification probabilities are reported on a common percentage scale. Mean misclassification averages individual-wave predictions using survey weights. Columns (5)–(6) add person-wave indicators for observations included in matching panel B but not C and in panel A but not B, using the Table 2 sample-construction rules. Analytical standard errors in parentheses use the survey-weighted sandwich covariance and delta method. Under stationarity, initial and steady-state employment coincide. An inconsistency may arise from response error, matching error, or both.

Appendix Table A2: Age- and education-inconsistency models: supplementary estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Indicators only		Indicators + large increments		Plus matching indicators	
	Stationary	Free α	Stationary	Free α	Stationary	Free α
<i>Employment-state transformations</i>						
Steady-state employment (%)	47.92** (0.09)	54.17** (0.62)	47.92** (0.09)	54.28** (0.62)	47.92** (0.09)	54.57** (0.63)
Initial minus steady employment (p.p.)	0.00 (fixed)	-6.56** (0.64)	0.00 (fixed)	-6.68** (0.65)	0.00 (fixed)	-6.98** (0.65)
<i>Misclassification by inconsistency pattern</i>						
No inconsistency (%)	2.25** (0.03)	2.26** (0.03)	2.27** (0.03)	2.27** (0.03)	1.98** (0.03)	1.98** (0.03)
Age inconsistency (%)	17.96** (0.26)	17.99** (0.26)	7.76** (0.37)	7.76** (0.36)	5.46** (0.28)	5.46** (0.28)
Age inconsistency, large (%)	17.96** (0.26)	17.99** (0.26)	21.46** (0.33)	21.50** (0.33)	15.83** (0.33)	15.86** (0.33)
Education inconsistency (%)	6.48** (0.12)	6.49** (0.12)	5.99** (0.18)	5.99** (0.18)	4.47** (0.15)	4.47** (0.15)
Education inconsistency, large (%)	6.48** (0.12)	6.49** (0.12)	6.58** (0.15)	6.60** (0.15)	4.82** (0.14)	4.83** (0.14)
Both inconsistencies (%)	31.94** (0.30)	31.99** (0.30)	17.24** (0.71)	17.24** (0.70)	11.31** (0.59)	11.30** (0.59)
Both inconsistencies, both large (%)	31.94** (0.30)	31.99** (0.30)	35.28** (0.35)	35.34** (0.35)	27.28** (0.50)	27.33** (0.50)
Panel B but not C, no inconsistency (%)	—	—	—	—	2.77** (0.06)	2.77** (0.06)
Panel A but not B, no inconsistency (%)	—	—	—	—	4.33** (0.09)	4.34** (0.09)
<i>Probability increments</i>						
Additional large age increment (p.p.)	0.00 (fixed)	0.00 (fixed)	13.70** (0.48)	13.73** (0.48)	10.37** (0.39)	10.40** (0.39)
Additional large education increment (p.p.)	0.00 (fixed)	0.00 (fixed)	0.59** (0.23)	0.60** (0.23)	0.35* (0.17)	0.36* (0.17)
Panel B-not-C increment (p.p.)	—	—	—	—	0.79** (0.05)	0.79** (0.05)
Panel A-not-B increment (p.p.)	—	—	—	—	2.35** (0.08)	2.36** (0.08)
<i>Logistic-link coefficients</i>						
δ_0 (intercept)	-3.0530** (0.0145)	-3.0526** (0.0145)	-3.0467** (0.0144)	-3.0463** (0.0144)	-3.1893** (0.0162)	-3.1888** (0.0161)
δ_1 (age inconsistency)	2.4742** (0.0247)	2.4766** (0.0247)	1.3525** (0.0562)	1.3520** (0.0561)	1.0904** (0.0558)	1.0891** (0.0557)
δ_2 (education inconsistency)	1.1487** (0.0220)	1.1502** (0.0220)	1.0521** (0.0340)	1.0523** (0.0339)	0.8695** (0.0349)	0.8690** (0.0348)
δ_3 (large age increment)	0.00 (fixed)	0.00 (fixed)	1.4089** (0.0610)	1.4120** (0.0609)	1.3297** (0.0597)	1.3329** (0.0596)
δ_4 (large education increment)	0.00 (fixed)	0.00 (fixed)	0.1077** (0.0414)	0.1099** (0.0413)	0.0826* (0.0401)	0.0846* (0.0400)
δ_5 (panel B but not C)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.3514** (0.0215)	0.3516** (0.0215)
δ_6 (panel A but not B)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.8339** (0.0239)	0.8358** (0.0239)
Log-likelihood	-333.651M	-333.601M	-333.352M	-333.301M	-332.881M	-332.826M
N	394,823	394,823	394,823	394,823	394,823	394,823
Stationarity imposed	Yes	No	Yes	No	Yes	No
Large-increment effects	No	No	Yes	Yes	Yes	Yes
Matching-quality indicators	No	No	No	No	Yes	Yes

Note: Probabilities and increments are percentages or percentage points; logit coefficients are untransformed. A large inconsistency lies at least two units beyond the admissible progression. Panel-membership probabilities set age and education inconsistencies to zero and vary one matching indicator at a time. Analytical standard errors use the survey-weighted sandwich covariance and delta method.

The additional reliability-group robustness table is generated as table_inconsistency_reliability.tex in EM-baseline-ext/output/tables.

Interpretation

- Baseline misclassification is 2.26%, rising to 17.99% with an age inconsistency, 6.49% with an education inconsistency, and 31.99% with both.
- Stationarity materially changes the entry and exit decomposition but barely changes the inconsistency-related misclassification probabilities. In the free model, initial employment is 6.56 percentage points below its implied steady-state rate.
- Allowing additional large-increment effects substantially sharpens the age gradient: predicted misclassification rises from about 7.76% for a mild age inconsistency to 21.50% for a large one. The corresponding education increase is only about 0.60 percentage points.
- In the augmented free-initial-employment model, predicted misclassification for a record with no age or education inconsistency is 1.98% when retained by both stricter panels, 2.77% when retained by B but not C, and 4.34% when retained by A but not B. Thus the matching-quality indicators capture a strong reliability gradient.
- Controlling for matching quality attenuates but does not eliminate the age/education gradients: predicted error is 5.46% for a mild and 15.86% for a large age inconsistency, and 4.47% and 4.83% for mild and large education inconsistencies. The mean error rate changes little, from 3.36% to 3.38%, and the free-model entry and exit rates move only from 2.60% and 2.19% to 2.57% and 2.14%.
- The relationship is not explained solely by different true transition rates among records with age or education inconsistencies.
- Larger age inconsistencies are much more informative than mild ones; inconsistency severity matters less for education.
- Age and education inconsistencies predict higher status misclassification, without distinguishing response errors from incorrect panel matches.

To do

- Check alternative nonlinear definitions of inconsistency extent and more flexible inconsistency-group transition rates.
- Bootstrap the age/education-inconsistency effects and repeat the analysis under stricter panel matching.

4. Observable transition heterogeneity: Table 4

- Repaired the covariate-model optimiser so every specification maximises the appropriate likelihood, uses five starts including a nested-model warm start, and dominates its nested no-error model.
- Added analytical standard errors and posterior risk-set and survey-weighted transition rates.
- Added tenure and time since work in Set 3.
- Added time-varying contract type and informal-sector status in Set 4 without restricting the sample to respondents observed in employment at every wave.
- Moved raw coefficients, the initial-state parameter, total churn, and hazard distributions to an appendix table.

Table 4: Risk-set and survey-weighted quarterly employment transitions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No miscl.				Symmetric			
	Set 1	Set 2	Set 3	Set 4	Set 1	Set 2	Set 3	Set 4
Entry rate (%)	8.14 (0.05)	8.14 (0.05)	8.14 (0.05)	8.14 (0.05)	3.62 (0.06)	3.77 (0.06)	4.66 (0.06)	4.87 (0.06)
Exit rate (%)	8.58 (0.05)	8.58 (0.05)	8.58 (0.05)	8.58 (0.05)	3.71 (0.06)	3.89 (0.06)	4.81 (0.06)	5.09 (0.06)
Misclassification rate (%)	—	—	—	—	2.58 (0.03)	2.50 (0.03)	2.34 (0.04)	2.18 (0.03)
Employment rate (%)	47.94 (0.09)	47.94 (0.09)	47.94 (0.09)	47.94 (0.09)	47.83 (0.09)	47.84 (0.09)	47.85 (0.09)	47.87 (0.09)
Log-likelihood	-343.8M	-342.6M	-337.1M	-336.2M	-338.2M	-337.5M	-333.2M	-332.7M
<i>N</i>	394,823	394,823	394,823	394,823	394,823	394,823	394,823	394,823
<i>Covariates included</i>								
Age, Age ² , Educ.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
+ Race, Gender	No	Yes	Yes	Yes	No	Yes	Yes	Yes
+ Tenure (exit), time since work (entry)	No	No	Yes	Yes	No	No	Yes	Yes
+ Contract type, informal sector (exit only)	No	No	No	Yes	No	No	No	Yes

Note: All specifications estimate the initial employment probability freely. Rates average predicted hazards using survey weights and posterior probabilities of belonging to the relevant origin-state risk set. Employment is the survey-weighted posterior employed share. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation. The approximation does not incorporate strata or PSU clustering.

Appendix Table A3: Covariate-model coefficients and predicted-hazard distributions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No miscl.				Symmetric			
	Set 1	Set 2	Set 3	Set 4	Set 1	Set 2	Set 3	Set 4
<i>Other derived model quantities</i>								
Initial employment rate (%)	47.93** (0.09)	47.93** (0.09)	47.93** (0.09)	47.93** (0.09)	47.80** (0.10)	47.82** (0.10)	47.84** (0.10)	47.87** (0.10)
Total employment churn (%)	8.35** (0.04)	8.35** (0.04)	8.35** (0.04)	8.35** (0.04)	3.66** (0.05)	3.83** (0.05)	4.73** (0.05)	4.98** (0.05)
<i>Entry equation: $\Phi(X_{it}\beta_0)$</i>								
Intercept	-1.3658** (0.0034)	-1.2464** (0.0053)	-0.8703** (0.0069)	-0.8707** (0.0069)	-1.7943** (0.0077)	-1.6189** (0.0093)	-1.0588** (0.0100)	-1.0416** (0.0097)
Age	1.3871** (0.0239)	1.4214** (0.0241)	1.3336** (0.0266)	1.3364** (0.0266)	1.9100** (0.0457)	1.9325** (0.0454)	1.4236** (0.0526)	1.4300** (0.0490)
Age ²	-1.2894** (0.0247)	-1.3104** (0.0249)	-1.2065** (0.0264)	-1.2092** (0.0264)	-1.8170** (0.0486)	-1.8232** (0.0481)	-1.3515** (0.0512)	-1.3505** (0.0480)
Education	0.0283** (0.0037)	0.0386** (0.0038)	0.0071 (0.0039)	0.0071 (0.0039)	0.0313** (0.0065)	0.0483** (0.0068)	-0.0186** (0.0065)	-0.0137* (0.0063)
Race: Coloured	—	0.1472** (0.0111)	0.0632** (0.0113)	0.0631** (0.0113)	—	0.1508** (0.0190)	0.0720** (0.0174)	0.0715** (0.0169)
Race: Indian	—	-0.1557** (0.0263)	-0.1097** (0.0269)	-0.1096** (0.0269)	—	-0.2941** (0.0571)	-0.2356** (0.0532)	-0.2390** (0.0514)
Race: White	—	-0.0678** (0.0207)	-0.0535* (0.0211)	-0.0532* (0.0211)	—	-0.1598** (0.0409)	-0.1423** (0.0378)	-0.1506** (0.0368)
Female	—	-0.2247** (0.0068)	-0.1600** (0.0069)	-0.1599** (0.0069)	—	-0.2948** (0.0120)	-0.2044** (0.0115)	-0.2015** (0.0111)
log(1 + tenure)	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
log(1 + timesincework)	—	—	-0.2455** (0.0043)	-0.2455** (0.0043)	—	—	-0.3101** (0.0066)	-0.3071** (0.0064)
Never worked	—	—	-0.5417** (0.0086)	-0.5413** (0.0086)	—	—	-0.8704** (0.0241)	-0.8352** (0.0212)
Tenure missing	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Permanent contract	—	—	—	0.0000 [fixed]	—	—	—	0.0000 [fixed]
Informal sector	—	—	—	0.0000 [fixed]	—	—	—	0.0000 [fixed]
<i>Employment-persistence equation: $\Phi(X_{it}\beta_1)$</i>								
Intercept	1.2840** (0.0037)	1.2772** (0.0053)	1.1032** (0.0058)	1.0437** (0.0085)	1.7297** (0.0079)	1.6890** (0.0103)	1.3743** (0.0098)	1.2854** (0.0130)
Age	0.8081** (0.0305)	0.9015** (0.0309)	0.6830** (0.0317)	0.6915** (0.0318)	0.9079** (0.0550)	1.0118** (0.0543)	0.6174** (0.0529)	0.6434** (0.0505)
Age ²	-0.5637** (0.0295)	-0.6602** (0.0298)	-0.5788** (0.0306)	-0.5799** (0.0307)	-0.5689** (0.0546)	-0.6765** (0.0536)	-0.4860** (0.0526)	-0.5117** (0.0499)
Education	0.2019** (0.0032)	0.1756** (0.0034)	0.1438** (0.0036)	0.1049** (0.0038)	0.2461** (0.0048)	0.2118** (0.0051)	0.1720** (0.0057)	0.1230** (0.0058)
Race: Coloured	—	0.0969** (0.0104)	0.0800** (0.0105)	0.0197 (0.0107)	—	0.1348** (0.0191)	0.0510** (0.0172)	-0.0168 (0.0168)
Race: Indian	—	0.3125** (0.0250)	0.2556** (0.0259)	0.2307** (0.0262)	—	0.6983** (0.0861)	0.5734** (0.0647)	0.5341** (0.0626)
Race: White	—	0.3926** (0.0168)	0.3432** (0.0174)	0.3083** (0.0175)	—	0.8114** (0.0669)	0.7096** (0.0518)	0.6342** (0.0470)
Female	—	-0.0973** (0.0070)	-0.0744** (0.0071)	-0.0845** (0.0072)	—	-0.1181** (0.0123)	-0.0548** (0.0119)	-0.0678** (0.0115)
log(1 + tenure)	—	—	0.3214** (0.0048)	0.2998** (0.0049)	—	—	0.5980** (0.0151)	0.5083** (0.0120)
log(1 + timesincework)	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Never worked	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Tenure missing	—	—	-0.0350 (1.0497)	-0.0398 (1.0311)	—	—	-0.0938 (1.2670)	-0.0993 (1.2935)
Permanent contract	—	—	—	0.2191** (0.0086)	—	—	—	0.2657** (0.0135)
Informal sector	—	—	—	-0.1943** (0.0094)	—	—	—	-0.2417** (0.0136)
<i>Distribution of predicted entry hazards</i>								
10th percentile	3.86	3.67	3.21	3.20	1.07	0.97	0.89	1.00
Median	8.17	7.86	5.99	5.99	3.37	3.29	2.00	2.21
90th percentile	12.41	13.59	17.45	17.44	6.54	7.62	13.46	13.82
<i>Distribution of predicted exit hazards</i>								
10th percentile	4.37	3.29	2.15	1.94	1.02	0.41	0.14	0.19
Median	7.56	7.69	7.14	6.78	2.75	3.00	2.28	2.51
90th percentile	14.56	14.84	17.09	18.05	7.64	8.40	13.53	14.29
<i>Set 4 discrete effects on the exit probability</i>								
Permanent contract	—	—	—	-3.28** (0.13)	—	—	—	-2.43** (0.13)
Informal sector	—	—	—	3.01** (0.16)	—	—	—	2.36** (0.15)

Note: Raw probit coefficients are reported. Tenure enters persistence only; time since work and never worked enter entry only. Contract type and sector are origin-wave characteristics and may change between transitions; their entry coefficients are fixed at zero because these attributes are not defined for non-employed origin states. Hazard percentiles use survey and posterior risk-set weights. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation; quantile SE are omitted.

Interpretation

- No-error entry and exit remain close to 8.1% and 8.6% across covariate sets.
- Symmetric-error estimates range from 3.62–4.87% for entry and 3.71–5.09% for exit; estimated misclassification falls from 2.58% in Set 1 to 2.18% in Set 4.
- In symmetric Set 4, predicted risks are highly dispersed: the weighted 10th–90th percentile range is 1.00–13.82% for entry and 0.19–14.29% for exit.
- Observable heterogeneity therefore explains some differences in transition risk but does not explain away the classification-error signal.
- Duration is economically important: entry falls with time since work and exit falls with tenure.
- Permanent contracts reduce exit and informal employment raises it. These are conditional associations, not causal estimates.
- The paper should distinguish heterogeneous true transition risk from the upward bias in observed transitions caused by classification error.

To do

- Allow the initial employment probability to depend on baseline covariates and check whether this changes the transition and error estimates.
- Test more flexible duration functions and confirm the main inference with a person-level bootstrap.

5. Higher-order state dependence: Table 5

- Added the comparable four-wave AR(1) benchmark with symmetric error and free initial employment.
- Added the same tenure and time-since-work controls used in Table 4.
- Allowed duration covariates to change across origin waves.
- Replaced approximate optimisation and inference with exact collapsed likelihood estimation and analytical standard errors.

Table 5: Four-wave AR(1) and AR(2) estimates

	AR(1)	AR(2)	AR(2), duration controls
Entry rate: nonemployed in both preceding waves	3.20 (0.04)	4.60 (0.06)	4.51 (0.07)
Entry rate: employed two waves ago	3.20 (0.04)	30.75 (0.47)	17.48 (0.97)
Exit rate: nonemployed two waves ago	3.13 (0.04)	31.60 (0.48)	19.16 (0.81)
Exit rate: employed in both preceding waves	3.13 (0.04)	4.38 (0.06)	4.27 (0.08)
Misclassification rate	2.70 (0.03)	0.98 (0.04)	1.49 (0.08)
Log likelihood	-468,443.8	-467,392.7	-462,205.1
N	317,437	317,437	317,437
Second employment lag	No	Yes	Yes
Duration controls	No	No	Yes

Notes. Entries are model-implied quarterly probabilities in percent; the duration-controlled entries are posterior risk-set and survey-weighted averages. Parentheses contain analytical standard errors. The AR(1) rate is repeated across the two corresponding AR(2) rows because the first-order model imposes equality across the second-lag state.

Every specification has one latent type, a constant symmetric misclassification probability, and free initial conditions. Duration controls comprise log tenure, log time since work, never-worked and missing-duration indicators; duration values vary across origin waves. The duration-controlled model's 317,437 observations collapse exactly to 37,031 likelihood cells.

Appendix Table A4: Four-wave AR(1) and AR(2) coefficients

Coefficient	AR(1)	AR(2)	AR(2), duration controls
Initial employment logit	-0.081 (0.004)	—	—
Initial-pair logit: 10	—	-2.677 (0.016)	-2.847 (0.031)
Initial-pair logit: 01	—	-2.674 (0.016)	-2.839 (0.030)
Initial-pair logit: 11	—	-0.081 (0.004)	-0.081 (0.004)
Entry: intercept	-1.852 (0.006)	-1.685 (0.006)	-1.207 (0.012)
Entry: employed two waves ago	—	1.182 (0.014)	0.811 (0.034)
Entry: log time since work	—	—	-0.215 (0.010)
Entry: never worked	—	—	-0.831 (0.028)
Entry: timegap missing	—	—	-0.132 (0.082)
Persistence: intercept	1.862 (0.006)	0.479 (0.013)	0.575 (0.025)
Persistence: employed two waves ago	—	1.230 (0.014)	0.903 (0.025)
Persistence: log tenure	—	—	0.431 (0.015)
Persistence: tenure missing	—	—	0.003 (0.719)
Misclassification logit	-2.862 (0.010)	-3.917 (0.042)	-3.486 (0.055)

Transition coefficients are on a common probit scale. The unrestricted AR(1) and no-duration AR(2) transition probabilities are re-expressed in their observationally equivalent probit parameterisations; the duration-controlled AR(2) is estimated directly by probit. Initial-state coefficients are logits, and the misclassification coefficient is $\text{logit}(2\pi)$ in all three models. Parentheses contain analytical sandwich/delta-method standard errors. Duration variables and their missingness indicators vary across origin waves. The information-matrix condition numbers are 14.6 for AR(1), 130.7 for AR(2), and 388,646 for duration-controlled AR(2); all three are locally full rank.

Interpretation

- The AR(1) model estimates quarterly entry and exit of 3.20% and 3.13%, with a 2.70% symmetric misclassification probability.
- Duration controls substantially reduce the large AR(2) transition differences but do not eliminate them.
- Entry and exit after two periods in the same state remain close to 4.5% and 4.3%; recent state reversals remain associated with much higher transition probabilities.
- Surviving AR(2) dependence may reflect unobserved duration, state dependence, latent heterogeneity, or within-quarter transitions; the current estimates do not distinguish these mechanisms.

To do

- Compare AR(1) and AR(2) using formal fit criteria and richer duration controls.
- With additional waves, test latent heterogeneity, wave-varying transitions, and correlated classification errors as competing explanations.

6. Unobserved heterogeneity: Table 6 and Appendix Table A5

- Re-estimated the two-type finite-mixture model by direct likelihood maximisation with multiple starts and audited local identification.
- Removed the underidentified three-wave FMM table: only the no-error model with free initial conditions is locally identified; symmetric error cannot be separated from two-type transition heterogeneity.
- Developed a four-wave alternative with observed covariates, duration controls, and class-specific misclassification intercepts.
- Compared all four specifications in one main table, including the nested no-misclassification model, and moved inconsistency-specific error rates to a secondary table.

Table 6. Comparison of all four-wave two-type FMM specifications

	No controls				Controls			
	No error		Common error		Common error		Type-specific error	
	Stable	Mobile	Stable	Mobile	Stable	Mobile	Stable	Mobile
Latent-class share (%)	75.65	24.35	92.13	7.87	89.63	10.37	58.17	41.83
	(1.00)	(1.00)	(0.37)	(0.37)	(1.26)	(1.26)	(3.05)	(3.05)
Entry rate (%)	0.46	30.46	2.36	42.16	3.20	27.30	1.33	11.98
	(0.18)	(0.75)	(0.07)	(1.15)	(0.12)	(2.49)	(0.15)	(0.64)
Exit rate (%)	0.004	33.63	2.21	47.04	3.01	26.87	0.75	11.18
	(0.240)	(0.77)	(0.08)	(1.16)	(0.16)	(1.57)	(0.20)	(0.65)
Misclassification rate (%)	0.00	0.00	1.35	1.35	1.74	1.98	0.68	4.60
	(fixed)	(fixed)	(0.03)	(0.03)	(0.03)	(0.03)	(0.09)	(0.18)
Initial employment rate (%)	47.73	48.17	47.74	47.95	47.38	50.59	44.61	52.33
	(0.18)	(0.45)	(0.13)	(0.74)	(0.22)	(1.75)	(0.58)	(0.56)
Log-likelihood	-451,874.52		-450,805.93		-428,777.56		-428,568.60	
<i>N</i>	308,240		308,240		308,240		308,240	

All estimates except the log likelihood and observation count are percentages; analytical survey-weighted sandwich standard errors, transformed by the delta method, are in parentheses. Entry and exit rates are posterior risk-set weighted in the controlled models. Posterior mean misclassification averages each respondent-wave's fitted posterior mismatch probability using survey weights and posterior class membership. Controls comprise observed covariates and duration controls. Both controlled models use common slopes and class-specific transition intercepts. The last model additionally makes the misclassification intercept class-specific; the age- and education-inconsistency slopes remain common. In the no-error model, all misclassification probabilities are fixed at zero. The no-error model has full probability-Jacobian rank, but weak curvature around its near-zero stable-class exit estimate warrants caution. Pattern-specific implied misclassification rates are reported separately in Appendix Table A5.

Appendix Table A5. Misclassification rates by inconsistency pattern

Inconsistency pattern	Common error: Stable	Common error: Mobile	Type-specific error: Stable	Type-specific error: Mobile	Mobile minus stable
None	0.72 (0.02)	0.72 (0.02)	0.14 (0.04)	2.37 (0.15)	2.22
Age	14.72 (0.31)	14.72 (0.31)	6.28 (0.98)	35.52 (1.05)	29.25
Education	3.73 (0.12)	3.73 (0.12)	0.89 (0.21)	11.83 (0.53)	10.94
Age and education	34.85 (0.40)	34.85 (0.40)	23.62 (1.97)	46.93 (0.34)	23.31

Estimates are misclassification rates in percent; analytical standard errors are in parentheses. Both specifications include observed covariates and duration controls. The common-error model has a common misclassification intercept; the type-specific model has separate stable- and mobile-class intercepts. Age- and education-inconsistency slopes are common across classes.

Appendix Table A5, Panel B. Preferred FMM coefficients and diagnostics

Quantity	Estimate	Analytical SE
Persistence: log tenure	0.650	0.015
Entry: log time since work	-0.325	0.007
Entry: never worked	-0.860	0.018
Persistence: permanent contract	0.399	0.016
Persistence: informal sector	-0.266	0.016
Mobile-minus-stable misclassification intercept	2.839	0.225
Wald statistic for common error intercept	12.64	—
Entry-risk peak age	38	—
Persistence peak age	53	—
Information-matrix rank	32/32	—
Minimum information eigenvalue	0.0000120	—
Information condition number	29,216	—
Maximum weighted score	0.0000032	—

Coefficients are probit-index effects except the logit-scale difference between the class-specific misclassification intercepts. The controlled sample contains 308,240 people and collapses exactly to 229,809 likelihood cells.

Interpretation

- The three-wave symmetric-error FMM is underidentified and should not be reported structurally; four waves are required to separate the two mechanisms.
- Imposing no misclassification pushes the stable-class exit estimate to essentially zero and attributes 30–34% quarterly transition rates to the mobile class. Allowing error materially improves fit and avoids this extreme decomposition.
- Four waves support a large stable/mobile distinction after controlling for observed characteristics and duration.
- Allowing the misclassification probability to differ by type makes the mobile class larger and less extreme, showing that the uncontrolled FMM mixed genuine mobility, observed heterogeneity, and heterogeneous classification error.
- In the preferred controlled model, the survey-weighted posterior mean misclassification rate is 0.68% in the stable class and 4.60% in the mobile class, a much sharper reliability contrast than the baseline probabilities alone convey.
- The type-specific probabilities describe joint latent classifications; they do not show that mobility causes reporting or matching error.
- The four-wave results are best presented as a replacement or appendix extension rather than as validation of the underidentified three-wave results.

To do

- Complete profile-likelihood, multistart, and simulation checks for the four-wave model’s identification and class separation.
- Test covariate-dependent class membership and, with a longer panel, whether latent types can be combined with AR(2) state dependence.

7. Endogenous duration dependence: Table 7

- Modelled transition hazards as functions of endogenous tenure and time since work, separately from the age/education-consistency analysis in Table 3.
- Repaired the tenure-contamination likelihood and optimisation diagnostics.
- Retained exact consistency between clean duration reports and latent employment histories.
- Integrated unavailable wrong-state duration clocks instead of imputing them.
- Replaced constant and power-law transition hazards with tail-safe piecewise hazards.
- Added contamination in reported time since work and tested local and joint alternatives.

Table 7: Tenure-contamination misclassification and transition rates

	Constant, exact	Power law, exact	Piecewise, exact	Independent error	Local error	Joint per-wave error
Entry rate	4.08 (0.01)	1.96 (0.01)	3.64 (0.04)	3.13 (0.03)	2.91 (0.03)	3.15 (0.03)
Exit rate	4.26 (0.01)	4.51 (0.01)	4.62 (0.01)	4.42 (0.01)	4.43 (0.01)	4.42 (0.01)
Misclassification rate	10.21 (0.03)	10.53 (0.03)	9.93 (0.03)	2.33 (0.02)	3.02 (0.02)	2.32 (0.02)
Initial employment rate	54.98 (0.08)	55.92 (0.08)	55.02 (0.09)	48.30 (0.08)	48.75 (0.08)	48.30 (0.08)
Tenure contamination rate	24.79 (0.07)	25.09 (0.07)	24.98 (0.07)	25.22 (0.06)	25.22 (0.06)	25.21 (0.06)
Timegap contamination rate	0.00 fixed	0.00 fixed	0.00 fixed	28.04 (0.11)	39.20 (0.15)	16.62 (0.07)
Log likelihood	-2,510,430.3	-2,475,676.8	-2,462,197.4	-2,330,075.3	-2,343,051.4	-2,320,294.9
N	402,923	402,923	402,923	402,923	402,923	402,923
Transition-hazard form	Constant	Power law	Piecewise	Piecewise	Piecewise	Piecewise
Tenure-error model	Independent marginal	Independent marginal	Independent marginal	Independent marginal	Independent marginal	Independent marginal
Timegap-error model	None (exact)	None (exact)	None (exact)	Independent marginal	Local category	Joint per-wave marginal

Notes. Rates are in percent and follow the common entry, exit, misclassification, and initial-employment order used in the other main tables; model-specific duration-error rates follow. Entry and exit are posterior risk-set and survey-weighted quarterly probabilities. Parentheses contain numerical observed-information/delta-method standard errors from the full covariance matrix. In every model, a contaminated tenure report is an independent draw from the marginal tenure distribution; otherwise it must follow the exact clock. ‘Exact’ means that no separate timegap-contamination process is allowed; status misclassification and tenure contamination remain allowed.

For the joint model, 16.62% is the per-report probability. It implies a 30.48% probability that at least one report in a two-report pair is contaminated.

Appendix Table A6, Panel A. Exact and alternative duration profiles

Duration interval	Exact exit hazard	Exact exit (%)	Independent- error exit (%)	Joint-model exit (%)
0–3 months	0.3420	8.35	7.79	7.79
3–12 months	0.3558	8.51	7.90	7.90
1–3 years	0.1757	4.30	4.23	4.23
3–5 years	0.1522	3.73	3.67	3.67
5+ years	0.1353	3.33	3.27	3.27

Duration interval	Exact entry hazard	Exact entry (%)	Independent- error entry (%)	Joint-model entry (%)
0–3 months	0.5060	9.15	7.66	7.88
3–12 months	0.2620	6.34	5.00	5.07
1–3 years	0.1098	2.71	2.27	2.27
3–5 years	0.1177	2.90	2.57	2.57
5+ years	0.1425	3.50	2.82	2.83

Appendix Table A6, Panel B. Auxiliary model comparisons

Quantity	Constant, exact	Power law, exact	Piecewise, exact
Parameters	5	7	13
AIC	5,020,870.7	4,951,367.6	4,924,420.8
Mean employment duration (months)	68.9	68.1	67.0
Mean nonemployment duration (months)	72.0	1,257.2	76.7
Comparable two-report timegap contamination (%)	0	0	0
Analytical SE	fixed	fixed	fixed

Quantity	Independent error	Local error	Joint per-wave error
Parameters	14	14	14
AIC	4,660,178.6	4,686,130.8	4,640,617.9
Mean employment duration (months)	69.6	69.6	69.6
Mean nonemployment duration (months)	95.6	102.2	94.8
Comparable two-report timegap contamination (%)	28.04	39.20	30.48
Analytical SE	0.11	0.15	0.12

Appendix Table A6, Panel C. Effect of integrating unavailable wrong-state clocks

Quantity	Imputed clock	Integrated clock
Entry rate (%)	2.08	1.96
Exit rate (%)	4.91	4.51
Misclassification rate (%)	10.35	10.53
Tenure contamination rate (%)	24.81	25.09
Exit duration slope	-0.4413	-0.3742
Entry duration slope	-0.9148	-0.8782
Mean nonemployment duration (months)	3,800	1,257

Appendix Table A6, Panel D. Timegap-contamination mechanism decomposition

	Clock-feasible	Reset-compatible	Other	Posterior
Interview interval	(%)	contradiction (%)	contradiction (%)	contamination (%)
Wave 1 to wave 2	84.42	2.82	12.76	28.20
Wave 2 to wave 3	84.95	2.98	12.07	27.87

In the power-law model, exit falls from 7.42% at zero tenure to 3.22% at ten years, while entry falls from 8.51% to 1.18%. Integrating rather than imputing wrong-state clocks affects 15.5% of the

posterior employment risk set and 4.5% of the nonemployment risk set. In the piecewise exact model, median employment and nonemployment durations are 35.3 and 51.4 months. For the independent-error model, clock-impossible shares are 15.6% and 15.0%, while average contaminated-branch probabilities among feasible reports are 14.96% and 15.09%. Roughly 10% of fitted contamination mass is reset-compatible, 43–45% comes from contradictions that a hidden reset cannot explain, and 45–46% is allocated among clock-feasible reports. Joint-model exit risks differ from the independent-error risks by less than 0.03 percentage points in every interval.

The independent-error information matrix has rank 14/14, condition number 45, and a sharply peaked likelihood around 28.04%; the boundary likelihood-ratio diagnostic is 264,244. Refined maximum scaled gradients are 2.0×10^{-4} for the exact piecewise fit, 2.1×10^{-6} for the local-error fit, and 2.5×10^{-6} for the joint model. These are numerical diagnostics, not substitutes for the standard errors reported in Table 7.

Interpretation

- Piecewise hazards fit much better than constant or power-law hazards and improve the weighted-likelihood AIC by about 26,947 relative to the power law, while avoiding its implausibly long nonemployment tail.
- Exit falls after the first year of tenure. Entry falls sharply with time since work and then levels off rather than approaching zero.
- Allowing timegap contamination reduces estimated status error from about 10% to 2–3%, while tenure contamination remains close to 25%.
- The joint per-wave model fits best and leaves transition rates and duration profiles almost unchanged relative to the simpler contamination model.
- Only about 3 percentage points of timegap contradictions can be explained by an unobserved within-quarter employment spell; hidden resets are not the main source of the contamination estimate.
- The results identify inconsistency among status, tenure, timegap, and linked histories. They do not determine whether the underlying cause is respondent error, matching error, or another data problem.

To do

- Add bootstrap or profile uncertainty and posterior-predictive checks for the joint status and duration distributions.
- Test alternative contamination densities, rounding and heaping rules, and tail cut-points while retaining transition-duration consistency.

8. No-error duration counterfactual (Appendix Table A7)

- Estimated flexible AR(1) and AR(2) transition models treating reported status changes as error-free.
- Simulated the duration clocks implied by those transitions.
- Compared simulated and observed tenure and timegap distributions, with transition-conditional graphs as the primary diagnostic.

Appendix Table A7: No-error duration-clock contradictions

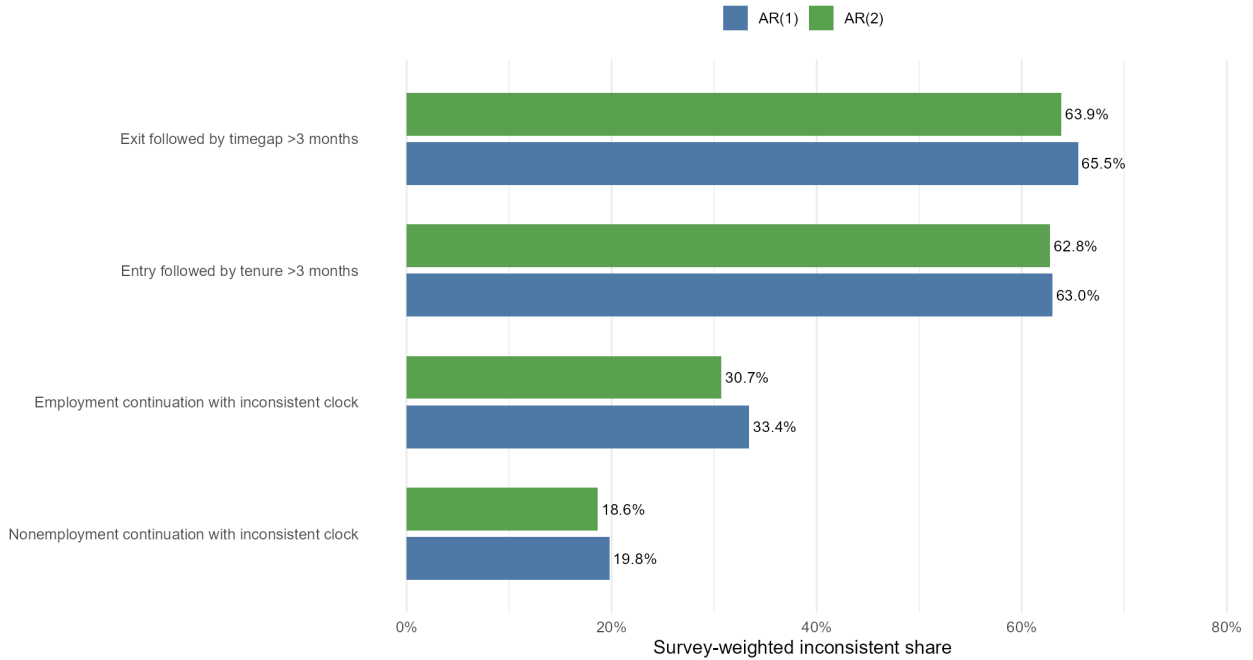
	AR(1)	AR(2)
Apparent entry followed by tenure over three months	63.0 (0.30)	62.8 (0.35)
Apparent exit followed by timegap over three months	65.5 (0.29)	63.9 (0.35)
Employment continuation with inconsistent tenure	33.4 (0.11)	30.7 (0.13)
Nonemployment continuation with unreachable timegap	19.8 (0.09)	18.6 (0.10)

Notes. Entries are survey-weighted shares in percent. Parentheses contain person-clustered linearized standard errors. Each no-error model allows flexible duration dependence in observed transitions; destination-wave durations are held out and used for these clock-consistency diagnostics.

Graph accompanying Panel A. Clock contradictions under AR(1) and AR(2)

Share of reported histories that violate a no-error spell clock

One-month tolerance is allowed for tenure increments; timegap uses reported category intervals



Appendix Table A7, Panel B. Observed transition benchmarks

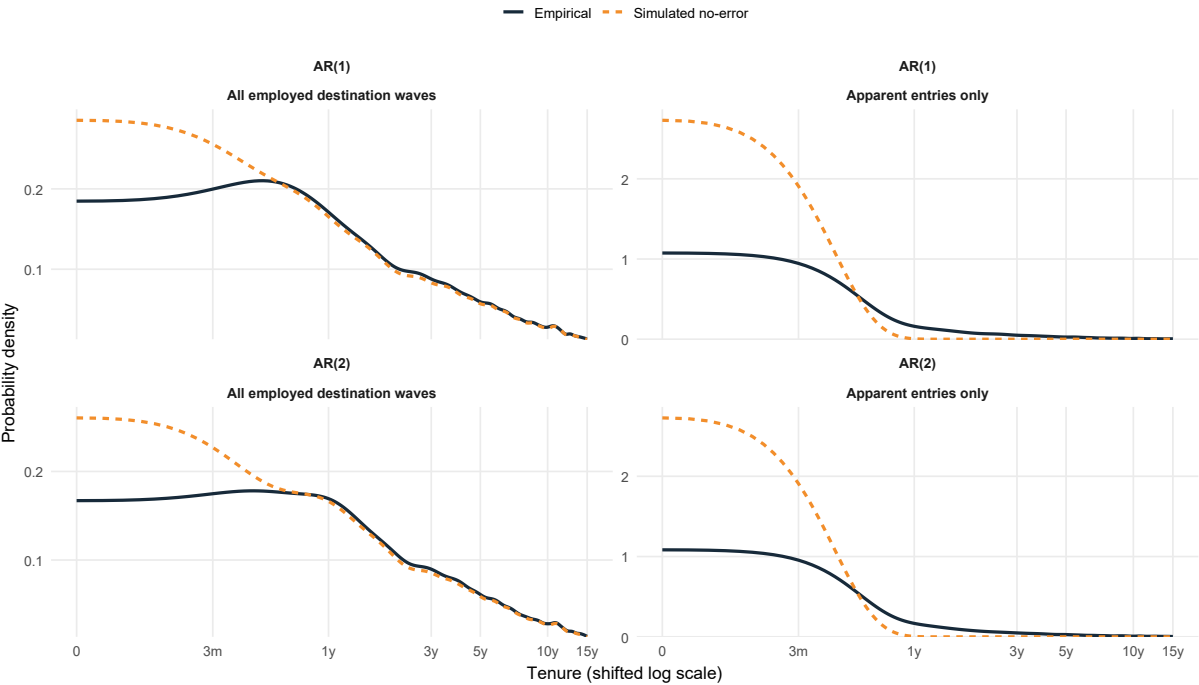
Rate	AR(1) (%)	AR(2) (%)
Entry rate	8.23	7.78
Exit rate	8.47	7.83
Admissible tenure after apparent entry	37.0	37.2

The flexible transition equations reproduce these observed entry and exit rates. Among apparent exits in the AR(1) analysis, 26.0% report that they have never worked, reinforcing the contradiction

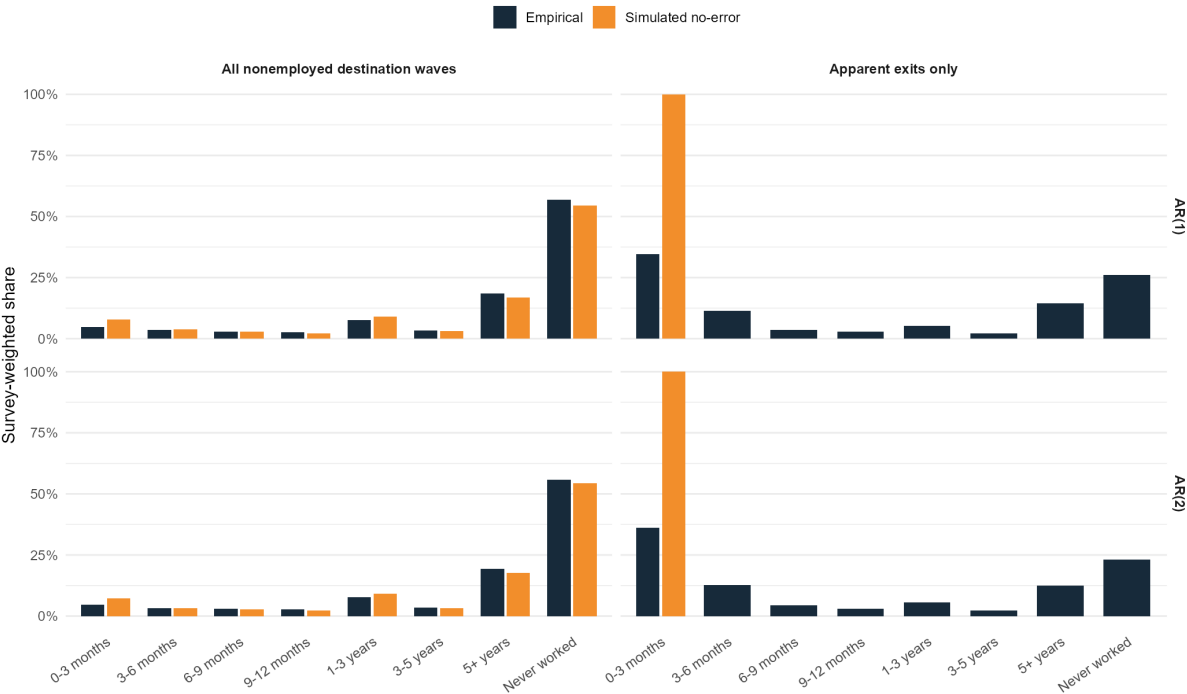
between status changes and the reported duration history.

Graphs accompanying Panel B. Empirical and simulated duration distributions

Empirical and simulated employment-tenure densities
Survey-weighted densities; log(tenure + one month) axis and common three-month bandwidth



Empirical and simulated time-since-employment distributions
Reported categories are compared with clocks propagated or reset by the no-error model



Interpretation

- Flexible no-error transition models reproduce average observed changes but cannot reconcile them with destination-wave duration reports.
- Most apparent entries and exits imply duration clocks that are impossible under a genuine transition during the interview interval.
- Marginal duration distributions obscure the problem because most respondents do not change reported state; transition-conditional comparisons are more persuasive.
- The diagnostic motivates jointly modelling status and duration consistency without claiming which reported variable or linked record is incorrect.

To do

- Add simulation uncertainty bands to the empirical-versus-implied duration graphs and report transition-conditional comparisons prominently.
- Repeat the counterfactual under alternative matching panels and initial-clock assumptions.

Reproducibility and remaining inference

- A result-by-result audit against `branch_changes_summary.Rmd` maps the baseline estimates to Table 1/A1, matching-rule results to Table 2, age/education-consistency results to Table 3/A2, covariate estimates and hazard distributions to Table 4/A3, AR(2) results to Table 5/A4, four-wave FMM results to Table 6/A5, tenure-contamination results to Table 7/A6, and the no-error duration benchmarks to A7. No numerical estimate reported in the long summary is intentionally omitted here; only detailed implementation narrative is abbreviated.
- Tables 1, 3, and 4 and Appendix Tables A1–A3 are rebuilt by `build_tables.R`.
- Table 2 is rebuilt by the matching-sensitivity script; the four-wave, tenure, and diagnostic scripts save the inputs for Tables 5–7 and Appendix Tables A4–A7 outside version control.
- Analytical standard errors are available for Tables 1–7 and their associated model estimates. Table 7 uses the full numerical observed-information matrix and delta method; Appendix Table A7 uses person-clustered linearization for its weighted descriptive shares.
- Table 7’s numerical observed-information errors should still be checked with a person-level bootstrap or profile likelihood before final submission.
- A person-level bootstrap remains the most useful common inference robustness check across the paper.